Type MS · Make an algorithm faster or work for any amount of data

Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 1 of 9 · find part b in this mark scheme extract

4(a)(i)	A1	Award **one** mark for: • (At the highest level) in the main scope/program/file (1)	Java and C#: allow answers that indicate inside namespace e.g. 'before main class'	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Make an algorithm faster or work for any amount of data

Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 2 of 9 · find part b in this mark scheme extract

4(a)(ii)	A2	Award **one** mark for: • Only from within the scope/subprogram/code block in which it is created/declared (1)		**1**

Question	mp	Answer	Additional guidance	Mark

Type MS · Make an algorithm faster or work for any amount of data

Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 3 of 9 · find part b in this mark scheme extract

4(b)	B1 B2	Award **two** marks for a linked explanation such as: • **Figure 4** executes fewer comparisons/tests (1), because (the use of nested ifs/else ifs/ means that) as soon as a test evaluates to true, all the remaining tests are skipped (1) • **Figure 5** executes more comparison/tests (1), because every if statement must be executed, even if any test before has already evaluated to true (1)		**2**

Type MS · Make an algorithm faster or work for any amount of data

Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 4 of 9 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Make an algorithm faster or work for any amount of data

Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 5 of 9 · find part b in this mark scheme extract

4(c)		Award **one** mark for each:		
	C1	Subprogram must attempt to use both pString and pNum to generate Key (1)	Award attempted use, even if not correct	
	C2	1D indexing used to access chars in string (1)	Can be anywhere in code (main or subprogram) Any method e.g. • pString[0]	
	C3	2 pairs of separated chars obtained (1)	First 2 **and** last 2 characters obtained from pString Any method Can be anywhere in code • pString[0] + pString[1] • pString[2] + pString[3] • pString[0:2] • pString[2:4]	
	C4	Key concatenated correctly (1)	Key must be correct Can be anywhere in code	
	C5	Subprogram must return their key to the calling line (1)	Do not award return without a value • Value must be variable/value intended to be key	
	C6	Complete line to call new subprogram (1)	Must be fully correct `myNewKey = genNewKey(myString, myNumber)`	**6**

Type MS · Make an algorithm faster or work for any amount of data

Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 6 of 9 · find part b in this mark scheme extract

C#

```
// Q04cFINISHED

namespace Q04cFINISHED
{
    class Q04cFINISHED
    {
        static String genNewKey (String pString, int pNum)
        {
            String newKey;

            // Write your code below this line
            newKey = pString.Substring (0, 2);            // First two chars
            newKey = newKey + pNum.ToString();            // Number
            newKey = newKey + pString.Substring (2, 2);   // Last two chars

            return (newKey);
        }

        static void Main(string[] args)
        {
            String myNewKey = "";
            int myNumber = 0;
            String myString = "";

            Console.WriteLine ("Enter a string:");
            myString = Console.ReadLine();
            Console.WriteLine ("Enter a whole number: ");
            myNumber = Convert.ToInt32(Console.ReadLine());

            if (myString.Length != 4)
            {
                Console.WriteLine ("String must be four characters");
            }
            else
            {
                Console.WriteLine ("Original: " + myString +
                        " " + myNumber.ToString());

                // Complete the call to the subprogram
                myNewKey = genNewKey (myString, myNumber);
                Console.WriteLine ("New: " + myNewKey);
            }
        }
    }
}

```

Type MS · Make an algorithm faster or work for any amount of data

Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 7 of 9 · find part b in this mark scheme extract

Java

```java
// Q04cFINISHED

import java.util.Scanner;

public class Q04cFINISHED
{
    static String genNewKey (String pString, int pNum)
    {
        String newKey;

        // Write your code below this line
        newKey = pString.substring(0,2);  // First two chars
        newKey = newKey + String.valueOf (pNum);              // Number
        newKey = newKey + pString.substring (2, 4);   // Last two chars

        return (newKey);
    }

    public static void main(String[] args)
    {
        String myNewKey = "";
        int myNumber = 0;
        String myString = "";
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a string:");
        myString = myKeyboard.nextLine();
        System.out.println ("Enter a whole number: ");
        myNumber = myKeyboard.nextInt();

        if (myString.length() != 4)
        {
            System.out.println ("String must be four characters");
        }
        else
        {
            System.out.println ("Original: " + myString +
                                " " + String.valueOf(myNumber));

            // Complete the call to the subprogram
            myNewKey = genNewKey (myString, myNumber);
            System.out.println ("New: " + myNewKey);
        }
    }
}

```

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Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 8 of 9 · find part b in this mark scheme extract

Python

```python
# Q04cFINISHED

def genNewKey (pString, pNum):
    newKey = ""         # Make new word here

    # Write your code below this line
    newKey = pString[0:2]               # First two chars
    newKey = newKey + str (pNum)        # Number
    newKey = newKey + pString[2:4]      # Last two chars
    return (newKey)

myString = input ("Enter a string: ")
myNumber = int (input ("Enter a whole number: "))

if (len (myString) != 4):
    print ("String must be four characters")
else:
    print ("Original: ", myString, myNumber)

    # Complete the call to the subprogram
    myNewKey = genNewKey (myString, myNumber)
    print ("New word:", myNewKey)

```

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Answer 1 of 4 · Question 4 part b · 2 marks

2024 JUN · Paper 02 · extract 9 of 9 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

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Answer 2 of 4 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 01 · extract 1 of 8 · find part b.ii in this mark scheme extract

6(a)	Indicative content provided Other solutions do exist and should be awarded		6

Aspect of Solution	Marks			
	0	1	2	3
Functionality	No awardable content	There are significant errors in logic, leading to an overall solution that is non-functional	There are minor errors in logic, leading to an overall solution that is not completely functional	There are no errors in logic, leading to an overall solution that is fully functional
Accuracy of Notation	No awardable content	Notation follows a broadly unrecognisable convention that is applied inconsistently, although aspects of it are discernible	Notation follows a recognisable convention which is broadly discernible but is applied inconsistently	Notation follows a recognisable convention and is applied consistently throughout

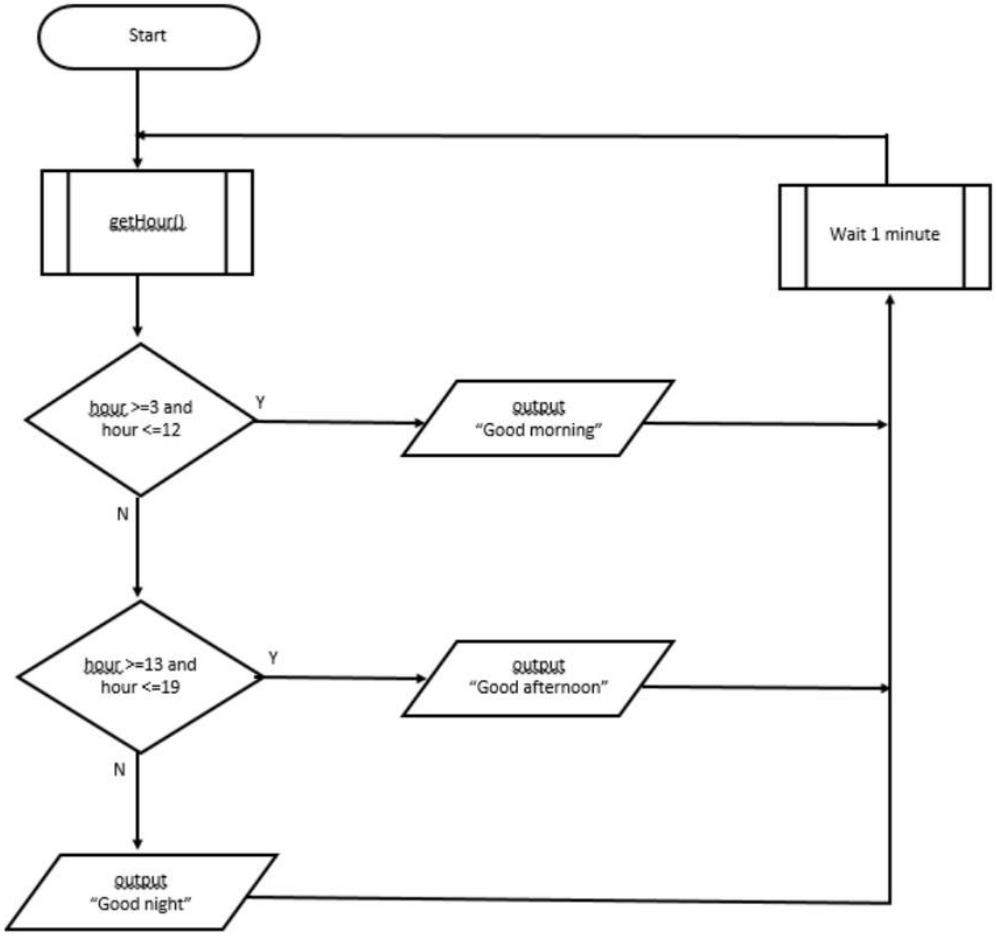
There are a maximum of 3 marks for functionality.
There are a maximum of 3 marks for accuracy of notation.
The marks for functionality and accuracy are awarded independently.

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Answer 2 of 4 · Question 6 part b.ii · 1 mark

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18

Type MS · Make an algorithm faster or work for any amount of data

Answer 2 of 4 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 01 · extract 3 of 8 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	• papaya needs changing (1) • lychee needs ordering (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(ii)	7 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(b)(iii)	Replace 8 with LENGTH(flavours)/LENGTH(volume) (1) or A new line 7: WHILE i < LENGTH(flavours) DO (1) or WHILE i < LENGTH(volume) DO (1)	Ignore case If new line is given award mark if logic is clear	1

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Type MS · Make an algorithm faster or work for any amount of data

Answer 2 of 4 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 01 · extract 4 of 8 · find part b.ii in this mark scheme extract

Question Number	Answer	Mark

20

Type MS · Make an algorithm faster or work for any amount of data

Answer 2 of 4 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 01 · extract 5 of 8 · find part b.ii in this mark scheme extract

6(c)	Indicative content: **Compiler** • One line of a compiled language maps to multiple executable instructions • Reads in a whole file and translates it at once • Produces an executable file • Executable file is portable between machines with the same architecture and operating systems • End-user cannot see the programming source code • Does not need an additional environment/software to run the code **Interpreter** • One line of a compiled language maps to multiple executable instructions • Reads, translates, and executes one line at a time • A special environment is needed to be installed on the user's machine to run the code; this is machine specific • Source code is portable to any machine with an interpreter which can run on it • End-user can see the programming source code **Similarities** • One line of a compiled language maps to multiple executable instructions/one to many relationship to machine code • Both are used to interpret high level programming languages • Both produce machine code from human-readable programming code **Differences** • Compiler reads in a whole file and translates it at once. Interpreter reads, translates and executes one line at a time • Compiler produces an executable file, interpreter does not • Executable files produced by a compiler will only execute on machines with the same architecture and operating systems. Source code produced by an interpreter is portable to any machine with an interpreter which can run on it • Compiled code does not require any additional software	6

Type MS · Make an algorithm faster or work for any amount of data

Answer 2 of 4 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 01 · extract 6 of 8 · find part b.ii in this mark scheme extract

	to execute the code. An interpreter needs a special environment to be installed on the user's machine to run the code; this is machine specific	

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Answer 2 of 4 · Question 6 part b.ii · 1 mark

2020 NOV · Paper 01 · extract 7 of 8 · find part b.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 2 of 4 · Question 6 part b.ii · 1 mark

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Type MS · Make an algorithm faster or work for any amount of data

Answer 3 of 4 · Question 6 part b.iii · 1 mark

2020 NOV · Paper 01 · extract 1 of 8 · find part b.iii in this mark scheme extract

6(a)	Indicative content provided Other solutions do exist and should be awarded		6

Aspect of Solution	Marks			
	0	1	2	3
Functionality	No awardable content	There are significant errors in logic, leading to an overall solution that is non-functional	There are minor errors in logic, leading to an overall solution that is not completely functional	There are no errors in logic, leading to an overall solution that is fully functional
Accuracy of Notation	No awardable content	Notation follows a broadly unrecognisable convention that is applied inconsistently, although aspects of it are discernible	Notation follows a recognisable convention which is broadly discernible but is applied inconsistently	Notation follows a recognisable convention and is applied consistently throughout

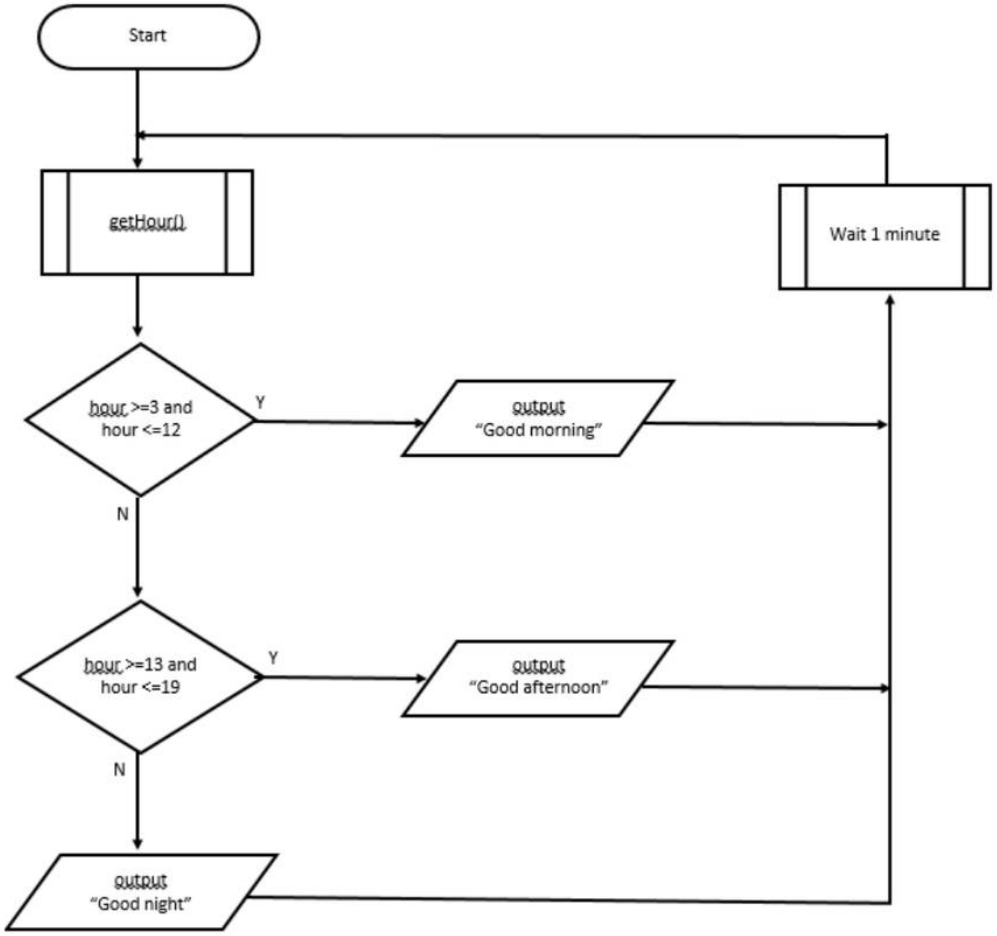
There are a maximum of 3 marks for functionality.
There are a maximum of 3 marks for accuracy of notation.
The marks for functionality and accuracy are awarded independently.

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Answer 3 of 4 · Question 6 part b.iii · 1 mark

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Type MS · Make an algorithm faster or work for any amount of data

Answer 3 of 4 · Question 6 part b.iii · 1 mark

2020 NOV · Paper 01 · extract 3 of 8 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	• papaya needs changing (1) • lychee needs ordering (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(ii)	7 (1)		1

Question Number	Answer	Additional Guidance	Mark
6(b)(iii)	Replace 8 with LENGTH(flavours)/LENGTH(volume) (1) or A new line 7: WHILE i < LENGTH(flavours) DO (1) or WHILE i < LENGTH(volume) DO (1)	Ignore case If new line is given award mark if logic is clear	1

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Type MS · Make an algorithm faster or work for any amount of data

Answer 3 of 4 · Question 6 part b.iii · 1 mark

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Question Number	Answer	Mark

20

Type MS · Make an algorithm faster or work for any amount of data

Answer 3 of 4 · Question 6 part b.iii · 1 mark

2020 NOV · Paper 01 · extract 5 of 8 · find part b.iii in this mark scheme extract

6(c)	Indicative content: **Compiler** • One line of a compiled language maps to multiple executable instructions • Reads in a whole file and translates it at once • Produces an executable file • Executable file is portable between machines with the same architecture and operating systems • End-user cannot see the programming source code • Does not need an additional environment/software to run the code **Interpreter** • One line of a compiled language maps to multiple executable instructions • Reads, translates, and executes one line at a time • A special environment is needed to be installed on the user's machine to run the code; this is machine specific • Source code is portable to any machine with an interpreter which can run on it • End-user can see the programming source code **Similarities** • One line of a compiled language maps to multiple executable instructions/one to many relationship to machine code • Both are used to interpret high level programming languages • Both produce machine code from human-readable programming code **Differences** • Compiler reads in a whole file and translates it at once. Interpreter reads, translates and executes one line at a time • Compiler produces an executable file, interpreter does not • Executable files produced by a compiler will only execute on machines with the same architecture and operating systems. Source code produced by an interpreter is portable to any machine with an interpreter which can run on it • Compiled code does not require any additional software	6

Type MS · Make an algorithm faster or work for any amount of data

Answer 3 of 4 · Question 6 part b.iii · 1 mark

2020 NOV · Paper 01 · extract 6 of 8 · find part b.iii in this mark scheme extract

	to execute the code. An interpreter needs a special environment to be installed on the user's machine to run the code; this is machine specific	

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Answer 3 of 4 · Question 6 part b.iii · 1 mark

2020 NOV · Paper 01 · extract 7 of 8 · find part b.iii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 3 of 4 · Question 6 part b.iii · 1 mark

2020 NOV · Paper 01 · extract 8 of 8 · find part b.iii in this mark scheme extract

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Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 1 of 11 · find part a in this mark scheme extract

5(a)	A1 A2	Award **two** marks for a linked explanation such as: • Replacing individual variables with an array (using a single name) (1) means that comparisons could be done in a loop (1) • The animals could be processed in a loop (1) because they could be stored in an array (under a single name) (1)		**2**

Question	mp	Answer	Additional guidance	Mark

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 2 of 11 · find part a in this mark scheme extract

5(b)		Award **one** mark for each of:	Ignore contents of fourth row if given.	
	B1	First row high index and low index (1) • 9, 0	Accept • 20, 10	
	B2	First row mid point (1) • 4	Accept • 15	
	B3	Second row (1) • 3, 0, 1	Accept • 14, 10, 12	
	B4	Third row (1) • 0, 0, 0	Accept • 10, 10, 10	**4**

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Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 3 of 11 · find part a in this mark scheme extract

Example: Using index

highIndex	lowIndex	midPoint
9	0	4
3	0	1
0	0	0

Example: Using Values

highIndex	lowIndex	midPoint
20	10	15
14	10	12
10	10	10

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 4 of 11 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 5 of 11 · find part a in this mark scheme extract

5(c)		Award **one** mark for each of:		
	C1	Opens the file for reading (1)		
	C2	Reads each line from the file (1)	Requires both read lines (prime, inside loop) for C# and Java. Requires a foreach for Python. Allow FT from C1	
	C3	Splits each line into individual strings (1)	Syntactically correct <anyString>.split(',') Does not need to be splitting correct line	
	C4	Individual values cast to int for processing (1)	Any method (index or foreach) allow float no FT allowed from C1 to C3 – must be full values between commas in file	
	C5	SubTotal += value **and** grandTotal += subTotal (1)	Any reasonable attempt at calculating both totals	
	C6	All given variable and constant names used throughout (1)	INPUT_FILE COMMA subTotal grandTotal	
	C7	Subtotals displayed / Grand total displayed (1)	Must be accurate as shown in QP Fig 8 Not hard-coded in any way	
	C8	The input file is closed, before exiting (1)	Syntactically correct Must be file name originally opened – FT from C1	**8**

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Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 6 of 11 · find part a in this mark scheme extract

Q	mp	Code Examples

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 7 of 11 · find part a in this mark scheme extract

C#

```
// Q05cFINISHED

using System.IO;

String inputFile = "Sales.txt";     // Output file name
String comma = ",";                 // Use as a constant

int subTotal = 0;               // Subtotal for each line
int grandTotal = 0;             // Running total
String line;                    // Line read from file

// Complete the code to open the file for reading
String fullPath = "C:\\Q05c";
inputFile = fullPath + "\\" + inputFile;
StreamReader theFile = new StreamReader (inputFile);

// Complete the code to read the first line of the input file
line = theFile.ReadLine ();

// Loop as long as the line is not null
while (line != null)
{
    // Complete the code to split the line into a set of five strings
    string[] stringSales = line.Split (comma);

    subTotal = 0;
    // Add code to sum up each value in the set of five strings
    foreach (string sale in stringSales)
    {
        subTotal = subTotal + Convert.ToInt32 (sale);
    }

    // Add code to display the subtotal for the line
    Console.WriteLine (Convert.ToString (subTotal));

    // Add code to calculate the running total
    grandTotal = grandTotal + subTotal;

    // Complete the code to read the next line of the file
    line = theFile.ReadLine ();
}

// Add code to display the total of all lines in the file
Console.WriteLine ("Grand total: " + Convert.ToString (grandTotal));

// Complete the code to close the opened file
theFile.Close ();

```

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 8 of 11 · find part a in this mark scheme extract

Java

```java
// Q05cFINISHED

import java.io.FileReader;
import java.io.BufferedReader;

public class Q05cFINISHED
{
    public static void main(String[] args) throws Exception
    {
        String inputFile = "Sales.txt";     // Output file name
        String comma = ",";                  // Use as a constant

        int subTotal = 0;                // Subtotal for each line
        int grandTotal = 0;              // Running total
        String line;                     // Line read from file

        // Complete the code to open the file for reading
        String full_path = "C:\\Q05c";
        inputFile = full_path + "\\" + inputFile;
        BufferedReader theFile = new BufferedReader (new FileReader (inputFile));

        // Complete the code to read the first line of the input file
        line = theFile.readLine();

        // Loop as long as the line is not null
        while (line != null)
        {
            // Complete the code to split the line into a set of five strings
            String[] stringSales = line.split (comma);

            subTotal = 0;
            // Add code to sum up each value in the set of five strings
            for (int i = 0; i < stringSales.length; i++)
            {
                subTotal = subTotal + Integer.parseInt (stringSales[i]);
            }

            // Add code to display the subtotal for the line
            System.out.println (String.valueOf (subTotal));

            // Add code to calculate the running total
            grandTotal = grandTotal + subTotal;

            // Complete the code to read the next line of the file
            line = theFile.readLine();
        }

        // Add code to display the total of all lines in the file
        System.out.println ("Grand total: " + String.valueOf (grandTotal));

        // Complete the code to close the opened file
        theFile.close ();
    }
}
```

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 9 of 11 · find part a in this mark scheme extract

Python

```python
# Q05cFINISHED

INPUT_FILE = "Sales.txt"    # Output file name
COMMA = ","                 # Use as a constant

subTotal = 0        # Subtotal for each line
grandTotal = 0      # Running total

# Complete the code to open the file for reading
theFile = open (INPUT_FILE, "r")

# Complete the code to read each line of the input file
for line in theFile:

    # Complete the code to split the line into a set of five strings
    stringSales = line.split (COMMA)

    subTotal = 0
    # Add code to sum up each value in the set of five strings
    for sale in stringSales:
        subTotal = subTotal + int (sale)

    # Add code to display the subtotal for the line
    print (str (subTotal))

    # Add code to calculate the running total
    grandTotal = grandTotal + subTotal

# Add code to display the total of all lines in the file
print ( "Grand total: " + str (grandTotal))

# Complete the code to close the opened file
theFile.close ()

```

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 10 of 11 · find part a in this mark scheme extract

Type MS · Make an algorithm faster or work for any amount of data

Answer 4 of 4 · Question 5 part a · 2 marks

2024 JUN · Paper 02 · extract 11 of 11 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark